Characterization of coherent flow structures in brain ventricles

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Abstract

The dynamic flow of cerebrospinal fluid (CSF) in brain ventricles exhibits flow features on several scales, both spatially and temporally. Most analysis of this complex flow and the accompanying transport has used instantaneous (Eulerian) flow variables. Such analysis makes understanding of unsteady transport challenging. Here, we analyze brain ventricular CSF flow both in a Eulerian sense and from the Lagrangian perspective – a time-integrated view of the flow. With geometries generated from imaging data, we model CSF flow in adult human and embryonic zebrafish brain ventricles. In the human brain we model flow governed by cardiovascular pulsations, CSF secretion and motile cilia. The flow driven by cardiovascular pulsations is derived from a damped linear elastic model of brain ventricle deformations, as a result of applying displacement boundary conditions derived from experimental data. In the zebrafish brain we consider flow driven solely by motile cilia. The tissue and flow models are implemented and solved with finite element methods. We use the resulting velocity fields to compute finite-time Lyapunov exponent (FTLE) fields and use these fields to characterize Lagrangian coherent structures, which can be approximated by ridges in the FTLE fields. These coherent structures demonstrate prominent flow features in the brain ventricles congruent with findings in experimental research. In the human brain ventricles, we also investigate the role of inertia by comparing flow models governed by the Navier-Stokes and the unsteady Stokes equations. Comparisons show that solving the unsteady Stokes equations is adequate to compute integrated flow variables like stroke volumes, but that the unsteady Stokes approximation fails to resolve intricate features of flow and advective transport that are present in the solution to the Navier-Stokes equations, features that could be important to elucidating transport.

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